
EDUCATION

- **School of the Gifted Young, University of Science and Technology of China** Hefei, China
Bachelor of Computer Science and Technology (Member of the Hua Xia Talent Program) Sept. 2024 – Jun. 2028
- **Current GPA:** 4.11/4.3 (Rank: 2/231)
- **Selected Courses:** Mathematical Principles and Algorithms of AI: 95; Linear Algebra(B1): 97; Mathematical Analysis(B1): 91; Computer Programming: 99; Data Structures A: 98; Graph Theory(H): 94.
- **Current research interests:** RL methods and applications for improving LLM alignment and multi-agent systems.
- **Cross-disciplinary potentials:** Strong background knowledge and keen interest in Chemistry. (e.g. **Organic Chemistry A1 (English): 99; Inorganic Chemistry I: 96.**)

RESEARCH EXPERIENCE

- **Personalized Generation of Children's Stories (Sept. 2025 - Mar. 2026(expected)):** Supervised by Prof. Xiangnan He @ USTC. Developed a personalized storytelling app by designing a multi-agent collaborative storytelling framework. One agent extracts user preferences, while the generator agent, finetuned for creating children's literature, generates a draft, which is handed to and given advice on by a critic agent. The agent framework is trained through reinforcement learning, with hybrid rewards from humans and LLMs.
- **Material Science Research Paper Comprehension Agent (2026.1 - Present):** Supervised by Prof. Wei Hu @ USTC. Based on FAISS indexing, I plan to build a vector database in the field of materials, combining Retrieval-Augmented Generation (RAG) mechanisms to enable consistency verification, thereby improving the accuracy and traceability of information extraction. At the knowledge level, I plan to model the relationships between materials, chemical processes, and properties as a heterogeneous graph structure, and use Relational Graph Convolutional Networks (R-GCN) for knowledge embedding learning to achieve incremental updates and logical reasoning of the knowledge graph.

PROJECTS

- **Mathematical Fundamentals and Algorithms of AI course project(2025):** Predicted molecular carcinogenicity on the NCI1 dataset. Investigated different types of Graph Neural Networks (GNN) including traditional GCNs and newer ones combined with optimal transport. Conducted experiments on the NCI1 dataset, and discovered that Batch Normalization offers an incremental boost in performance.
- **Wordle difficulty classification - Group Leader (2025):** Inspired by the Problem C of 2023 Mathematical Contest in Modeling, I collaborated with two other students to analyze how Wordle difficulties changed after 2023. I analyzed the data gathered from X(Twitter) and extracted key features based on statistical analysis. I then constructed a random forest model to predict the attempt distribution, optimized with Bayesian search. Results demonstrated that compared with those in 2023, Wordle puzzles in 2024 are "harder", with an increase in expected average attempts of +0.3.

SKILLS

- **Python :** Proficiency in **PyTorch** and transformer libraries including peft, trl, transformers etc. (experience in implementing Supervised Finetuning and Reinforcement Learning), and GNN related libraries including PyG; **Scientific computation and data processing:** Familiar with pandas, sklearn, numpy etc.
- **C++:** Competent in data structure and algorithms.
- **Others:** Git, L^AT_EX, Typst, System Verilog.
- **English:** Professional Fluency(TOEFL 112, GRE Verbal 164 + Quantitative 170 + Analytical Writing 4).

HONORS AND AWARDS

- 2025.12 **National Scholarship** (Top < 3% at School of the Gifted Young);
- 2025.11 Second Prize (Provincial) , in The 14th Chinese Mathematics Competitions (Non-Math Major);
- 2025.11 **Silver Prize**, 2025 ICPC Asia Wuhan Regional Contest (Team Leader);

- 2025.10 **Silver Prize**, 2025 ICPC Asia Chengdu Regional Contest (Team Leader);
- 2025.09 Hua Xia Talent Program in Computer Science and Technology Scholarship;
- 2025.05 First Prize (Provincial), Blue Bridge Cup Python Coding Contest;
- 2025.01 Language and Academic Standardized Test Scholarships at School of the Gifted Young;
- 2024.12 Bronze Prize, USTC Freshman Scholarship.